

Practical week-3

TASK-1

Develop a C program using fork() that creates a parent and child process. Display the Process ID(PID),Parent Process ID(PPID),and process states at different stages of execution

```
GNU nano 8.7.1 task1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main(){
    pid_t pid;
    printf("Parent process started\n");
    printf("Parent PID :%d\n",getpid());
    printf("Parent PPID :%d\n",getppid());

    pid=fork();

    if(pid<0){
        perror("fork failed");
        return 1;
    }
    else if(pid==0){
        printf("\n--- Child Process ---\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());
        printf("Child State: Running\n");

        sleep(5);

        printf("Child State: Terminating\n");
        exit(0);
    }

    else
    {
        // Parent process
    }
}
```

[Read 46 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```

else
{
    // Parent process
    printf("\n--- Parent Process ---\n");
    printf("Parent PID  : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    printf("Parent State: Running\n");

    printf("Parent State: Waiting for child...\n");
    wait(NULL);
    printf("Parent State: Child terminated\n");
    printf("Parent State: Terminating\n");
}

return 0;
}

```

[^]G Help [^]O Write Out [^]F Where Is [^]K Cut [^]T Execute
[^]X Exit [^]R Read File [^]\ Replace [^]U Paste [^]J Justify

```

← sirichandana@siri: ~
sirichandana@siri:~$ nano task1.c
sirichandana@siri:~$ gcc task1.c -o task1
sirichandana@siri:~$ ./task1
Parent process started
Parent PID  : 656
Parent PPID : 356

--- Parent Process ---
Parent PID  : 656
Parent PPID : 356
Parent State: Running
Parent State: Waiting for child...

--- Child Process ---
Child PID   : 657
Child PPID  : 656
Child State: Running
Child State: Terminating
Parent State: Child terminated
Parent State: Terminating
sirichandana@siri:~$ |

```

TASK-2:

To observe the different process states-Ready, Running, Waiting, and Terminated using Linux monitoring tools such as ps, top, and /proc.

Sol:

```
GNU nano 8.7.1 process_states.c
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Process started!\n");
printf("PID: %d\n",getpid());
printf("Running state...\n");
sleep(2);

printf("Waiting/Sleeping state...\n");
sleep(10);

printf("Process terminating...\n");

return 0;
}

[ Read 15 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
← sirichandana@siri: -
sirichandana@siri:~$ gcc process_states.c -o process_states
sirichandana@siri:~$ ./process_states
Process started!
PID: 832
Running state...
Waiting/Sleeping state...
Process terminating...
sirichandana@siri:~$ |
```

```
sirichandana@siri:~$ ps -o pid,ppid,state,cmd -p 832
```

```
PID    PPID  S  CMD
832    356  S  ./process_states
```

```
sirichandana@siri:~$ ./oop
```

```
sirichandana@siri:~$ top
```

```
top - 13:56:56 up 51 min,  1 user,  load average: 0.00, 0.02, 0.00
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  3758.6 total,  3113.5 free,   462.1 used,   258.0 buff/cache
MiB Swap:  1024.0 total,  1024.0 free,    00.0 used,  3296.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
832	sirichandana	20	0	24164	15472	11632	S	0.0	0.4	0:00.00	process_states
1	root	20	0	3180	2200	2072	S	0.0	0.1	0:01.23	systemd
407	sirichandana	20	0	22340	13272	10992	S	0.0	0.3	0:00.35	gnome-terminal-
902	sirichandana	20	0	8164	5996	3896	R	0.0	0.2	0:00.13	top

```
sirichandana@siri:~$ ./process_states &
```

```
[3] 1261
```

```
Process started!
```

```
PID: 1261
```

```
Running state...
```

```
sirichandana@siri:~$ Process terminating...
```

```
Waiting/Sleeping state...
```

```
sirichandana@siri:~$ ps -o pid,ppid,state,cmd -p 1261
```

```
PID    PPID  S  CMD
1261    356  S  ./process_states
```

```
sirichandana@siri:~$ 0
```

```
[2]+  Done                  ./process_states
```

```
sirichandana@siri:~$ |
```